

Celeste Guimapi

IMT Atlantique · LS2N (STACK team) · Nantes, France
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PROFILE	PhD student at IMT Atlantique (LS2N, STACK team) working on the energy efficiency of cloud applications. My current work focuses on enforcing strict energy budgets on serverless platforms, implemented on Apache OpenWhisk and evaluated on the Grid'5000 testbed.	
EDUCATION	PhD in Computer Science	Jan 2025 – present
	IMT Atlantique · LS2N, STACK team · Nantes, France <i>Energy Efficiency of Applications in the Cloud.</i> Supervisors: Jean-Marc Menaud (Professor) and Remous-Aris Koutsiamanis (Senior Lecturer).	
	Master's Degree in Computer Science	2019 – 2024
	École Nationale Polytechnique · Yaoundé, Cameroon	
RESEARCH EXPERIENCE	Strict energy budgets for serverless platforms	2025 – present
	PhD research · IMT Atlantique - Design and implementation of a system that enforces strict energy budgets on serverless function executions while keeping multi-step applications consistent. - Implementation on Apache OpenWhisk (Go, Python); experimental evaluation on Grid'5000. Article in preparation.	
	Research Intern	Apr – Aug 2024
	IMT Atlantique · Nantes, France - Energy consumption of sorting algorithms as a function of input disorder, on Grid'5000 (Python, perf, Matplotlib) — presented at COMPAS 2024. - IoT system collecting energy metrics from a smart plug (Python, Mosquitto MQTT, Zigbee MQTT). - Energy impact of parallel and distributed computing and of the programming language on Raspberry Pi devices (C, Java, Python, OpenMP, MPICH).	
TEACHING	Smart Grid Project	Apr – May 2025
	IMT Atlantique · first year (L1) · lab sessions, 17 h	

Étude énergétique des algorithmes de tri en fonction du désordre dans les données à trier

Celeste Guimapi, Remous-Aris Koutsiamanis, Jean-Marc Menaud

Conférence francophone d'informatique en Parallélisme, Architecture et Système

HAL →

OTHER
EXPERIENCE

Data Scientist Intern

Jul – Sep 2023

Boissons du Cameroun · Cameroon

- Requirements gathering with department heads for software supporting control of the brewing process.
- Machine-learning model for diagnostic and prescriptive fault analysis (Python, MySQL, Power BI).

Software projects

Full-stack web, desktop and mobile applications

Ticketing and revenue monitoring for a gaming room (Next.js, Electron, React Native, PostgreSQL, WebSockets); real-estate search with interactive maps (Angular, Leaflet, Express.js); membership and payments management (React, Express.js, MySQL).

CONTINUING
EDUCATION

IBM DevOps and Software Engineering Professional Certificate

In progress

IBM · Coursera

Introduction to Object-Oriented Programming (in Java)

Dec 2021

EPFL · Coursera

Introduction to Programming (in Java)

Nov 2021

EPFL · Coursera

SKILLS

Programming

Python · Go · C · Java · SQL · JavaScript / TypeScript

Systems & research infrastructure

Apache OpenWhisk · Docker · Linux · Grid'5000 · perf · Git
Also familiar with Knative, OpenFaaS and Hadoop.

Software engineering

Next.js · React Native · Angular · Express.js · Flask · PostgreSQL · MySQL

LANGUAGES

French (native) · English (B2)

OTHER ACTIVITIES

Founder of **DeepTutoring**, a home-tutoring group for students in preparatory classes in Cameroon.